TIC 219752822

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        32.310021 d
  Epoch (t0)    2830.1448 BTJD
  Duration      0.1700 d
  Depth         0.050 %  (497 ppm)
  SNR           13.7

INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      11  (gate >= 3)
  Robust depth ratio     0.96  (gate >= 0.50)
  Robust depth sigma     9.6  (gate >= 7.0)
  Odd/even z-score       2.14  (consistent)
  Secondary eclipse      not detected
```

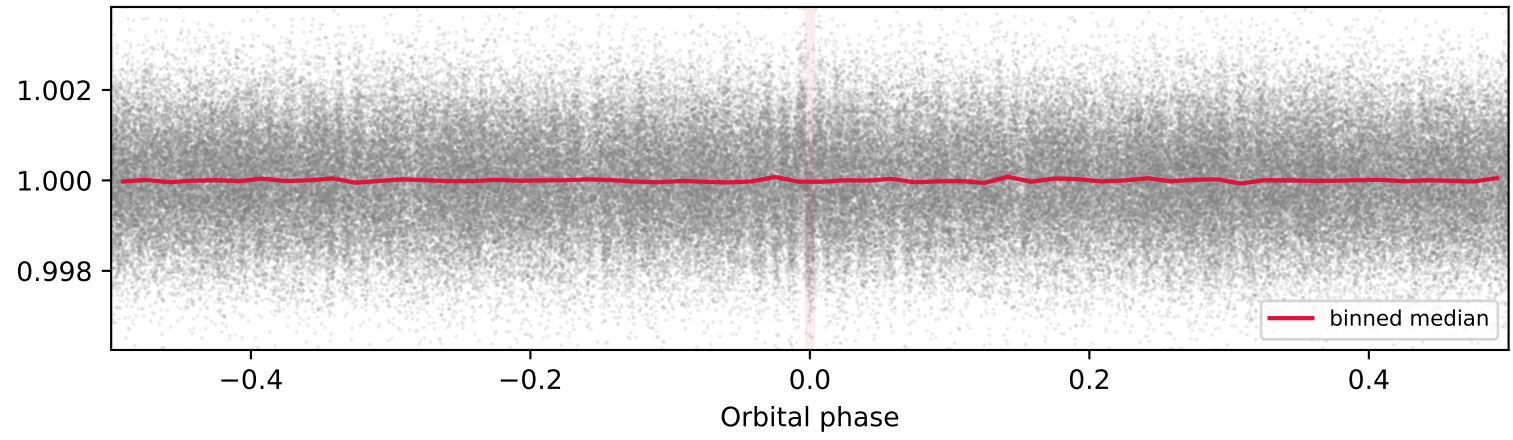
```
CATALOG CROSS-MATCH (novelty)

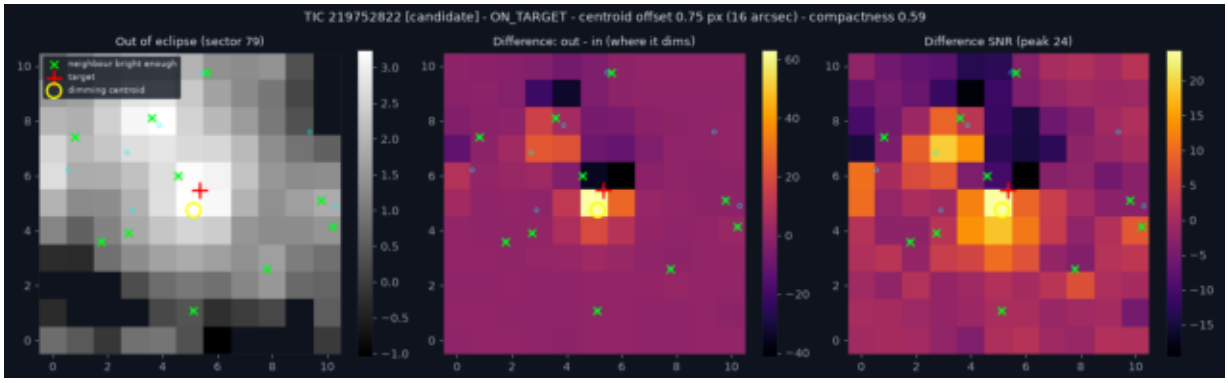
  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 79    light-curve dip in this sector: 6.3 sigma    t0 refined by -0.0125 phase
  Difference-image peak SNR   24.46   (gate >= 5.0)      Compactness   0.59   (gate >= 0.50)
  Dimming centroid offset     0.75 px (16 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 1400919732
    1.37 px away, dTmag +8.19 - farther from the dimming than the target
```

Phase-folded light curve (shaded = BLS eclipse window)





No caveats - clean detection on every check.